

```
#include <Servo.h>

// Motor A (RIGHT) connections
int enA = 3;

int in1 = 4;

int in2 = 5;

// Motor B (LEFT) connections
int enB = 8;

int in3 = 6;

int in4 = 7;

//Payload connection
int servoPin = 9;

int angle = 0;

Servo myServo;

void setup() {

    // Set all the motor control pins to outputs
    pinMode(enA, OUTPUT);
    pinMode(enB, OUTPUT);
    pinMode(in1, OUTPUT);
    pinMode(in2, OUTPUT);
    pinMode(in3, OUTPUT);
    pinMode(in4, OUTPUT);

    // Turn off motors - Initial state
    digitalWrite(in1, LOW);
    digitalWrite(in2, LOW);
```

```
digitalWrite(in3, LOW);  
digitalWrite(in4, LOW);  
  
// Attach the servo to the pin  
myServo.attach(servoPin);  
}
```

```
//Function Execution  
void loop() {  
    delay(10000);  
    drive10Feet();  
    delay(1000);  
    leftTurn90Degrees();  
    delay(1000);  
    drive10Feet();  
    delay(1000);  
    leftTurn90Degrees();  
    delay(1000);  
    drive10Feet();  
    delay(1000);  
    leftTurn90Degrees();  
    delay(1000);  
    drive10Feet();  
    delay(1000);  
    //Calling Payload Drop
```

```
dropPayload();  
delay(1000);  
}
```

```
//Mission 1-A 10 Ft. Square
```

```
//Drive 10 Ft. in a straight line at 1 ft/s
```

```
void drive10Feet(){
```

```
    // Turn on motors
```

```
    digitalWrite(in1, HIGH);
```

```
    digitalWrite(in2, LOW);
```

```
    digitalWrite(in3, LOW);
```

```
    digitalWrite(in4, HIGH);
```

```
    // Speed analog values for motors, change for testing
```

```
    // RIGHT MOTOR SPEED
```

```
    int straightSpeedA = 103;
```

```
    // LEFT MOTOR SPEED
```

```
    int straightSpeedB = 100;
```

```
    analogWrite(enA, straightSpeedA);
```

```
    analogWrite(enB, straightSpeedB);
```

```
    //for (int i = 64; i < straightSpeedA; i++) {
```

```
        //analogWrite(enA, i);
```

```
        //delay(5);
```

```

//}

//for (int i = 64; i < straightSpeedB; i++) {
    //analogWrite(enB, i);
    //delay(5);
//}

//Run function for 10 ft
delay(2000);

//for (int i = straightSpeedA; i >= 0; i=i-10) {
    //analogWrite(enA, i);
    //delay(20);
//}

//for (int i = straightSpeedB; i >= 0; i=i-10) {
    //analogWrite(enB, i);
    //delay(20);
//}

// Now turn off motors
digitalWrite(in1, LOW);
digitalWrite(in2, LOW);
digitalWrite(in3, LOW);
digitalWrite(in4, LOW);

}

//Perform a 90 degree turn in 5 seconds

```

```
void leftTurn90Degrees() {  
    // Turn on motors  
    digitalWrite(in1, LOW);  
    digitalWrite(in2, HIGH);  
    digitalWrite(in3, LOW);  
    digitalWrite(in4, HIGH);  
  
    // Speed analog values for motors, change for testing  
    int turnSpeedA = 128;  
    int turnSpeedB = 128;  
    analogWrite(enA, turnSpeedA);  
    analogWrite(enB, turnSpeedB);  
  
    // Run function for 5 seconds  
    delay(320);  
  
    // Now turn off motors  
    digitalWrite(in1, LOW);  
    digitalWrite(in2, LOW);  
    digitalWrite(in3, LOW);  
    digitalWrite(in4, LOW);  
  
}  
  
// Payload Drop Function  
void dropPayload() {
```

```
// Sweep the servo from 0 to 180 degrees
for(angle = 0; angle <= 110; angle += 22) {
  myServo.write(angle); // Set the servo position
  Serial.print("Angle: ");
  Serial.println(angle);
  delay (5);
  // Wait 15ms for the servo to reach the position
}

delay (1000);

// Sweep back from 180 to 0 degrees
for(angle = 110; angle >= 0; angle -= 22) {
  myServo.write(angle);
  Serial.print("Angle: ");
  Serial.println(angle);
  delay(5);
}

delay (1000);
}
```